UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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Contents

List of Figures

List of Tables

Chapter 1

Experiment: The Large Hadron Collider (LHC)

LHC is located at CERN (European Organization for Nuclear Research) located at Franco-Swiss border near Geneva, Switzerland and is most powerful setup of its kind to probe resarches in particle physics. From its design, it is capable enough to accelerate and then collide two proton beams [1] at center of mass energy upto 14 TeV at peak luminosity of $\mathscr{L} = 10^{34}\mathrm{cm}^2\mathrm{s}^{-1}$.It was constructed from 1998 to 2008, in a 3.8 meters wide tunnel of 27 Kilometers circumference at a depth which varies from 75-150 meters underground. This tunnel was previously used for LEP collider which was constructed from 1983 to 1988. Two counter rotating proton beams (anti-proton beam is disfavored to use) in two rings are used where the beam pipes are kept at ultra-high vacuum just to preserve the beams from possible extra collisions with molecules of gas. To orient the beams in alongside the tunnel superconducting (cryo) dipole magnets are used as sketched in 2.1.

1.1 LHC beam operations

Beams of protons are inserted to the LHC experiment at injector. Protons are obtained by removing electrons using high electric field. Injector is connected to Linac2, proton synchrotron booster, proton synchrotron and the super proton synchrotron booster (PSB) in which the proton beams pass and with a subsequent gain in energy at each step (50 MeV to 450 GeV from Linac2 to PSB). From the PSB, beam is then passed to LHC tunnel where beams accelerates until each beam attains the

[1] LHC also does collision experiments of Heavy(Pb) ions at 2.8 TeVper nucleon. Details are beyond the scope of the dissertaton

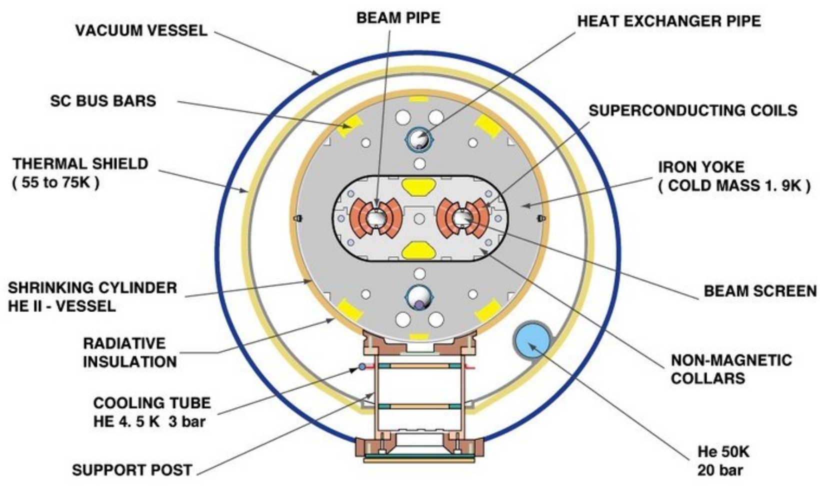

FIGURE 1.1: Cross section of the LHC cryodipole

energy of 6.5 TeVfor the subsequent collisions to be recorded at dedicated experiments or detectors of LHC. The schematic diagram of LHC accelerating mechanism is given in the Fig. 2.2

For the first time, beam of protons was injected into LHC on September 10, 2008. The initial test to circulate the beams in clockwise and counterclockwise throughout the collider was also successfully completed by the same day. On 19 September 2008, due to an electric defect magnet quench was triggered about a hundred bending magnets resulting in several damages. After the repair, in November 2009, LHC became world's strongest particle accelerator after having achieved 1.18 TeV energy each beam thus defeating the Tevatron's highest achieved energy of 0.98 TeV. In 2010, LHC kept boasting the beam energies and reached center of mass energy up to 7 TeV (3.5 TeV each beam). In 2011 and then in 2012, LHC consistently delivered data at 7 and 8 TeV energies respectively. During 2013-14 , LHC machine was shutdown to do major parts upgrade on the LHC machine and detectors installed on it. In 2015, upgraded LHC started again at 13 TeV close to full stipulated operating energies (14 TeV). After successfully delivering proton-proton collisions data during Run 2 till the end of 2018, LHC machine is shutdown for Long Shutdown 2 (LS2) upgrades and is expected be operational again for Run3 in 2021 after several upgrades.

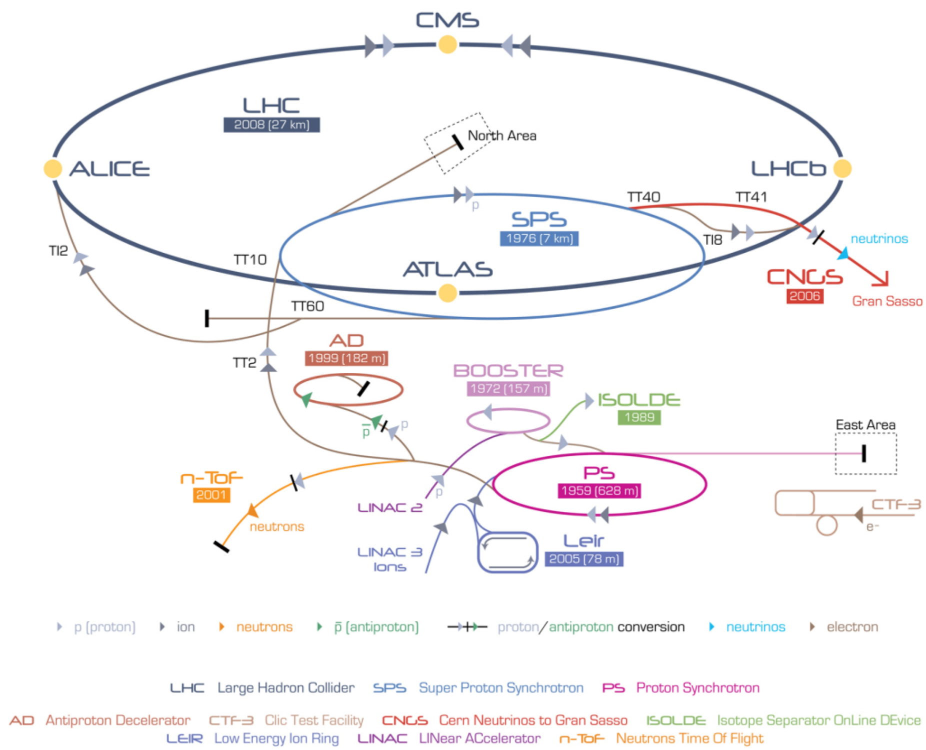

FIGURE 1.2: Accelerating mechanism of LHC experiment at CERN

1.1.1 Machine Layout and Performance

Basic layout of LHC is shown in Figure 2.2 [**Reference28**]. The LHC is designed to have two separate rings containing clockwise and counterclockwise proton beams, equipped with separate magnetic fields and vacuum chambers. These two rings only intersect each other at the points where detectors are installed. The ATLAS and CMS, two general purpose detectors of LHC, are installed opposite to each other (at P1 and P5 respectively). The other two important detectors, ALICE and LHCb, are installed (at P2 and P8 respectively) as shown in the same Figure 2.2 which are designed to study heavy ion Physics and B Physics (b quark) respectively. Two protons beams are injected in the vertical plane in both rings close to point 2 and point 8. In order to keep the beam alongside the tunnel during the process of accelerating, magnetic force is applied through diploe magnet made of $NbTi$ superconducting material. Apart from that, quadrupole magnet is used to focus the beam. Once each beam attains targeted energy i.e. 7 TeV, they are made collided at differents spots of LHC tunnel where different experiments are installed.

LHC design goal is to test SM Physics to ultimate precision and hunt for any kind of new physics up to 14 TeV. In collision process, number of events (N) per unit time (called event rate) for a certain process produced at LHC are given as

$$\frac{dN}{dt} = L \times \sigma_{process} \tag{1.1}$$

where $\sigma_{process}$ being the cross section of the process and L is the instantaneous machine luminosity. Integrating over time gives the total number of events in the particular process,

$$N = \int Ldt \times \sigma_{process} = \mathscr{L}\sigma_{process} \tag{1.2}$$

Here $\mathscr{L}$ represents *integrated* machine luminosity. All possible processes' cross sections produced during proton-(anti) proton collisions at LHC machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$ are shown in Figure 2.3. It is obvious from the diagram that an increase in machine luminosity and center-of-mass energies would enable us to get more statistics of Higgs boson. The Nominal luminosity for CMS and ATLAS experiment is $\mathscr{L}$ = 10^{34}cm^{2}s^{-1}, and nominal center of mass energy of LHC is $\sqrt{s} = 14$ TeV.

Nominal Luminosity in terms of Beam Parameters

The machine luminosity can be expressed in terms of beam parameters as:

$$\mathscr{L} = \frac{\gamma_r N_b^2 f_{rev} k_b}{4\pi\epsilon_n\beta^*}F \tag{1.3}$$

here γ_r represents relativistic factor; however, N_b is bunch intensity, k_b being number of bunches in a beam; f_{rev} is the revolution frequency, β^* represents the beta function at the collisions point and F is the luminosity depletion factor due to cross angle at interaction point (IP). This factor can be written as below

$$F = 1 + (\frac{\theta_c\theta_X}{2\sigma^*})^2 \tag{1.4}$$

where θ_c, θ_z and σ^* are crossing angle at point of interaction, root mean square (RMS) length of bunch and σ^* the transverse RMS size of beam at IP. In above expression of luminosity, some parameters can be used to increased the luminosity at LHC; however, some of them cannot be used e.g. f_{rev} which is fixed by perimeter

TABLE 1.1: Parameters' values for LHC operations.

Parameter	Value
Circumference	26659 m
Center of mass energy ($\sqrt{s}$)	14 TeV
Momentum at collision	7 TeV/s
Luminosity (nominal)	$10^{34} cm^2 s^{-1}$
Bunch spacing	25 ns
Minimum distance between bunches	7 m
No. of bunches in ring	2808
No. of protons in a bunch	1.15×10^{11}
Beta function at the collision point(β^*)	0.55 m
Transverse RMS size of beam at impact point(σ^*)	16.3 μm
Peak magnetic field of dipole	8.33 T
Beam lifetime	10 h
Current in main dipole	11800 A
Stored energy in magnets	11 GJ
No.of dipoles	1232
No.of quadrupoles	858
Minimum distance between bunches	7 m
Operating temperature of dipole magnet	1.9 K

of LHC tunnel and protons' speed in the beam. The expression above is valid for two circular beams with same parameters.

The two major experiment of LHC, the CMS [**Reference30**] and the ATLAS [**reference31**], are designed to operate at luminosity of $\mathscr{L} = 10^{34} cm^2 s^{-1}$. The summary of the parameters to obtain this nominal luminosity is summarized in Table 2.1.

By the end of the Run 2, LHC has delivered 150 $\text{fb}^{-1}$of data. However, to fully explore the SM, particulary the Higgs sector, and even beyond SM for the rare processes, physicists are really in need of sufficient statistics (i.e. higher luminosity). High luminosity LHC is, therefore, planned to achieve a peak luminosity of 5 $\times 10^{34} cm^2 s^{-1}$ - 10 $\times 10^{34} cm^2 s^{-1}$.

1.2 Data Taking from 2015 to 2018 (Full Run 2)

After successful data taking during Run 1, LHC machine was shutdown for two years in 2013 and 2014 for major upgrades in LHC machine and detectors. LHC started again in 2015 at center-of-mass energies to 13 TeV(close to its design

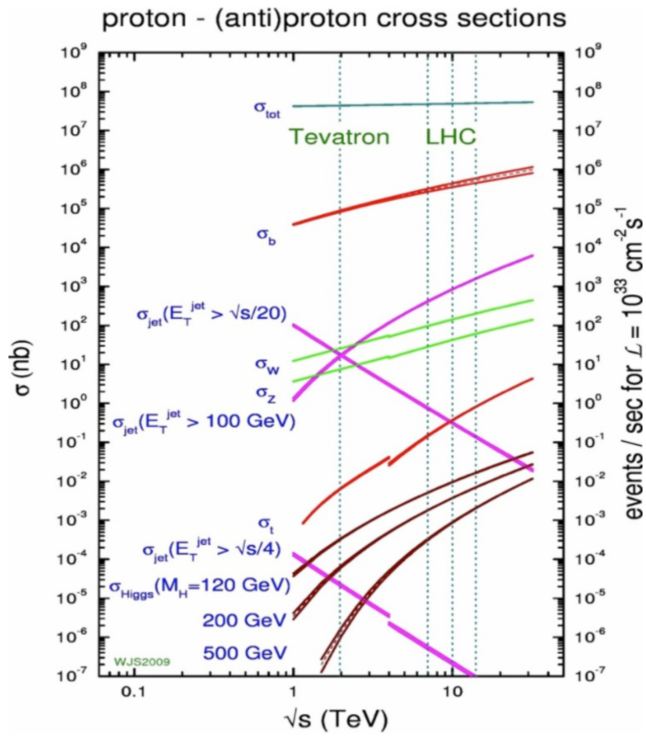

FIGURE 1.3: Cross section values for different process corresponding to machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$.

value i.e 14 TeV). LHC delivered 4.2 fb^{-1}during 2015 at 13 TeV center of mass energy and subsequently, it delivered 41.0 fb^{-1}, 49.8 $\mathrm{fb}^{-1}$and 67.9 fb^{-1}during the years 2016, 2017 and 2018 as shown in the Figure 2.4(a), 2.4(b), 2.4(c) and 2.4(d) respectively. In the figures we can also see the recorded data by the CMS experiment from LHC delivered data in different run periods.

1.3 Worldwide LHC Computing Grid

There is a dire demand of the four experiments of LHC (CMS, ATLAS, ALICE, LHCb) for a facility to store the prompt data obtained from LHC collisions. Further to process and analyse the huge amount of data to extract physics results is also challenging. LHC produces about 15 petabyte data each year duing its operation. To cope with this huge amount of data, setting a multi-tiered computing infrastructure was approved by CERN council in 2002. This is known as Worldwide LHC Computing Grid (WLCG) and which links around 170 computing centers in

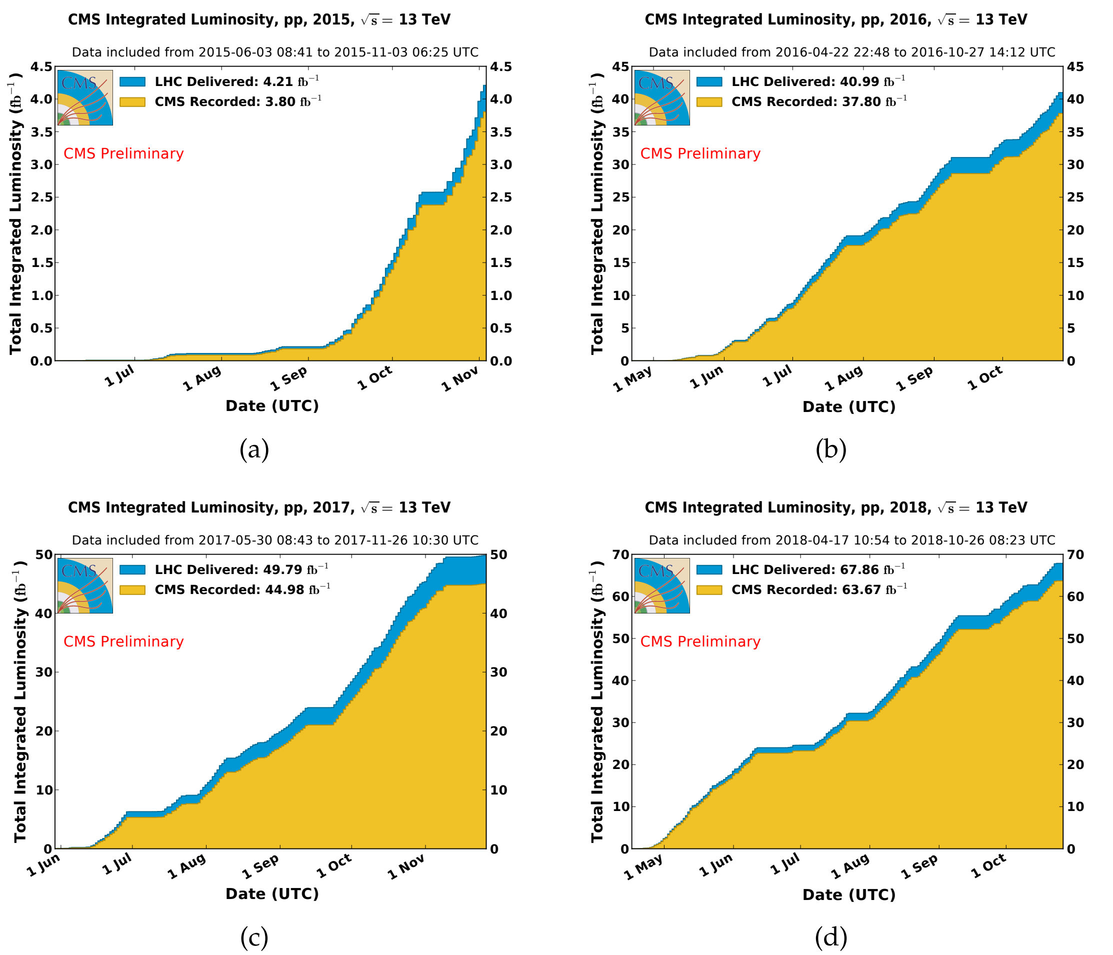

FIGURE 1.4: Integrated luminosities of LHC delivered (blue) and CMS recorded (yellow) data for year 2015(a), year 2016(b), year 2017(c) and year 2018(d).

more than 40 countries. WLCG is mainly tasked for providing infrastructure of organized computing facilities to the four main experiments of CERN. To fulfill the goals, it is connected with around 150 thousand cores of processors with storage capicity of 50 petabytes on the disks **[Fisk:2010jd]**.

1.3.1 3-Tier Architecture

Computing models at WLCG are mainly based on 3 tier levels. Tier-0 facility is situated at CERN. It receives prompt data (RAW data) produced from LHC collisions as a primary backup and does the reconstruction of the events and store in the form of prompt-RECO dta. It then exports the RAW data to to large computing centers at Tier-1 where the events are set to be reconstructed again with

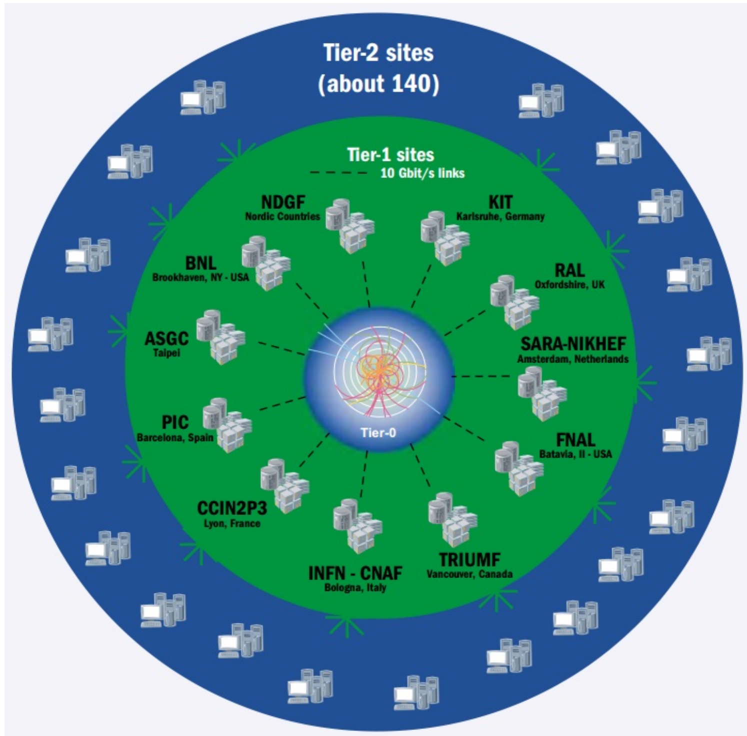

FIGURE 1.5: A sketch of WLCG Tier architecture.

refined calibrations and at each Tier-1 site, a detailed reconstruction of RAW data is performed to get AOD(Analysis Object Data) version of data which most compact form of data. It also serves to distribute the AOD data to Tier-2 sites where the data can be used for end-user analysis specific reprocessing and simulations for all experiments.

Schematic diagram of WLCG Tier system is shown in the Figure 2.5.

1.4 LHC upgradation during Long shutdown 2 (LS2)

After successfully delivering data during its Run2 as described in section 2.2, LHC is passing through its long shutdown 2 period. During this period, several upgardes are ongoing which can be regarded as preparations for High Luminosity LHC later on (HL-LHC). However, main upgrades in this regards will take place during LS3 after Run 3 period of LHC. One is the replacement of Linac2 with the Linac4. It is about 90 meter in length that took around 10 years to build it.It would be capable to accelerate negative hydrogen ions (H^-) for beam energy upto 160 MeV and will enable beam to double its intensity Ref. **??**. Besides, the individual experiments of LHC, i.e. CMS, ATLAS, ALICE and LHCb, have also planned certain upgarades during this LS2 period **??**.

ATLAS is builing muon small wheels to be integrated with the trigger system. For high pileup LAr electronics are being upgraded.

CMS is changing the photodetectors (HPD) of barrel and endcap HCAL to the silicon photo-multipliers (SiPM). Also the beam pipes made with stainless steel are being replaced with Aluminium composed material. Also, based on damage from radiation in endcaps of HCAL, scintillating tiles would be changed. Furthermore, for HL-LHC preprations, CSC muon chambers are being upgraded with new front end (FE) elecronics.

ALICE is making several upgrades for Inner Tracking System (ITS), Time Projection Chamber (TPS), Central Trigger Processor, Data Acquisition (DAQ) High Level Trigger (HLT), Muon Forward Tracker (MFT) and so on.

LHCb is improving its readout from 1 MHz to 40 MHz with trigger based fully on softwares by replaceing FE electronics and new computing infrastructure.

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